the Ranger are both deep in the cylindrical part of Gargantua's warped space (Figure 28.1), they are both dragged circumferentially by Gargantua's whirling space with almost the same angular velocity (the same orbital period). So as seen by Brand, in her orbiting reference frame, the Ranger drops away from the *Endurance* almost straight downward toward the horizon (Figure 28.1). That's what the movie depicts.

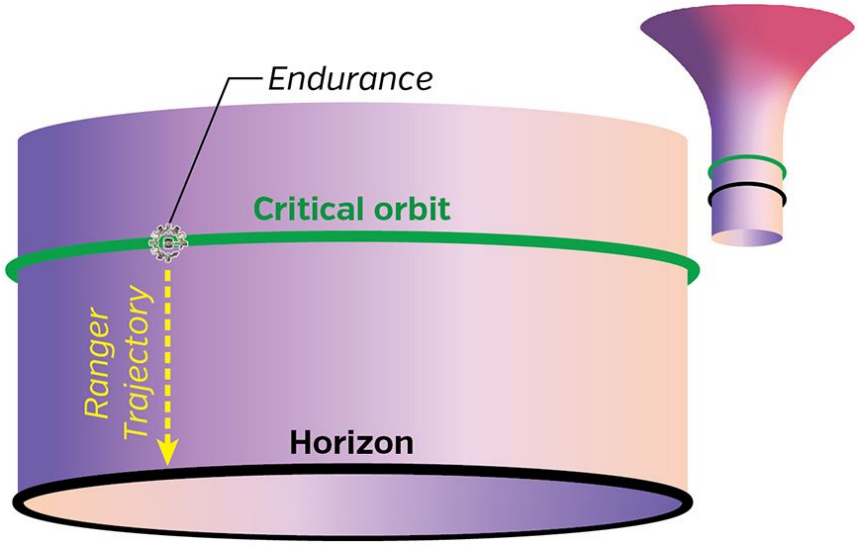


Fig. 28.1. The Ranger's trajectory through Gargantua's warped space, as seen in the *Endurance*'s orbiting reference frame. The *Endurance* is drawn far larger than it should be, so you can see it. *Inset*: A larger portion of Gargantua's warped space. *[Image of the* Endurance *is from* Interstellar.*]*

As Brand watches the Ranger approach the horizon, she must see time on the Ranger slow and then freeze relative to her time, Einstein's laws say. This has several consequences: She sees the Ranger slow its downward motion and then freeze just above the horizon. She sees light from the Ranger shift to longer and longer wavelengths (lower and lower frequencies, becoming redder and redder), until the Ranger turns completely black and unobservable. And bits of information that Cooper transmits to Brand one second apart as measured by his time on